T3 general-quartic benchmark atlas

Frozen 16 models x 3 scenarios | read-only presentation of campaign_frozen.csv

SOURCE INTEGRITY: SHA-256 B3B9FE0EF2F3730AE49504B51CDD7AED4D4232E6812CBBFE0D16D78CD23ED3AF
Accepted conditional: 20 (18 original-boundary + 2 modified-boundary); poor fits: 6; screened out: 22.
All five oscillation observables and scalar masses below come directly from the frozen CSV. Never interpret a screened result as an evaluated oscillation failure.

Scientific status

• Candidate/conditional general-quartic comparison, not a full general-potential or fully sequential matched EFT.
• Additional renormalisable UV/EFT1 scalar quartics are used where exported. Holomorphic quadratic extensions, full general-scalar Wilson matching, and physical Matchete/RGBeta normalisation certification are absent.
• Four extreme hypercharge cases use legacy common-scale C5 matching sources. B(-1)/C(-1) are imported, independently validated historical conditional fits; their running-UV cases used modified topology-coupling boundaries.
• Earlier Reports/output analytical equations, RGE dashboards and sensitivity ovals are retained as historical references, not recalculated general-quartic figures.
• No complete frozen 48-study RGE trajectories or local sensitivity covariance/scan outputs are in the campaign table. They are not invented in this atlas.

Source and scope

Dataset: output/studies/general_scalar_extended/bulk_48/bulk48_v1/campaign_frozen.csv
Additional quartic registry status: GeneratedCandidateBasisNotWolframVerified
The exact three scenario names are reproduced in the following pages. E-class lambdaT3 inputs are half-normalised (2.5e-9 / 0.25) as frozen.

Model representations and analytical references

Representation tuple is (SU(2) dimension of S1, S2, F); alpha is the model hypercharge parameter. These are not complete field quantum-number tables.

Model	(dS1,dS2,dF)	alpha	Registered extra quartics	Analytical source folder
A_-4	(1, 3, 2)	-4	0	Reports/output/models/T3-A-m4
A_-2	(1, 3, 2)	-2	5	Reports/output/models/T3-A-m2
A_+0	(1, 3, 2)	0	5	Reports/output/models/T3-A-p0
B_-3	(2, 2, 1)	-3	1	Reports/output/models/T3-B-m3
B_-1	(2, 2, 1)	-1	7	Reports/output/models/T3-B-m1
B_+1	(2, 2, 1)	1	1	Reports/output/models/T3-B-p1
C_-3	(2, 2, 3)	-3	1	Reports/output/models/T3-C-m3
C_-1	(2, 2, 3)	-1	7	Reports/output/models/T3-C-m1
C_+1	(2, 2, 3)	1	1	Reports/output/models/T3-C-p1
D_-2	(3, 1, 2)	-2	5	Reports/output/models/T3-D-m2
D_+0	(3, 1, 2)	0	5	Reports/output/models/T3-D-p0
D_+2	(3, 1, 2)	2	0	Reports/output/models/T3-D-p2
E_-4	(3, 3, 2)	-4	0	Reports/output/models/T3-E-m4
E_-2	(3, 3, 2)	-2	6	Reports/output/models/T3-E-m2
E_+0	(3, 3, 2)	0	6	Reports/output/models/T3-E-p0
E_+2	(3, 3, 2)	2	0	Reports/output/models/T3-E-p2

Analytical inventory

Each model-specific source folder contains UV and EFT1 Lagrangian, RGE and group-factor LaTeX, plus EFT2/Weinberg-stage files. Existing master PDFs/TeX are in Reports/output/analytical/. Generated additional-quartic symbolic sources are in RGE/general/generated/. These analytical files have not been relabelled as 48-study equations.

16 x 3 screening matrix

A = accepted original boundary; A* = accepted modified boundary; P = poor fit; X = no mass-safe fit.

Model	Small / frozen	Large / frozen	Small / running
A_-4	X	X	X
A_-2	P	A	X
A_+0	P	A	X
B_-3	X	X	X
B_-1	A	A	A*
B_+1	A	A	X
C_-3	X	X	X
C_-1	A	A	A*
C_+1	A	A	X
D_-2	P	A	X
D_+0	P	A	X
D_+2	A	A	X
E_-4	X	X	X
E_-2	P	A	X
E_+0	P	A	X
E_+2	A	A	X

Fit quality by scenario

Small topology quartic / frozen UV: 6 accepted; 6 poor; 4 screened out.
Large topology quartic / frozen UV: 12 accepted; 0 poor; 4 screened out.
Small topology quartic / running UV: 2 accepted; 0 poor; 14 screened out.

Small topology quartic / frozen UV

Saved low-energy endpoints; masses at muS (TeV). Missing entries are explicitly unavailable. B/C existing imported fits are not new bulk optimisations.

Model	status	chi2	mS1	mS2	lambdaT3
A_-4	Screened	unavailable	—	—	5e-09
A_-2	Poor	2.13681e+12	1.0541	1.1498	5e-09
A_+0	Poor	5.22338e+17	1.0685	1.1355	5e-09
B_-3	Screened	unavailable	—	—	5e-09
B_-1	Accepted	3.21545e-12	1.0164	1.1639	5e-09
B_+1	Accepted	4.17868e-06	1.044	1.1639	5e-09
C_-3	Screened	unavailable	—	—	5e-09
C_-1	Accepted	2.04541e-17	1.0164	1.1639	5e-09
C_+1	Accepted	4.17868e-06	1.044	1.1639	5e-09
D_-2	Poor	6.84988e+11	0.99189	1.2246	5e-09
D_+0	Poor	8.86451e+11	1.0043	1.208	5e-09
D_+2	Accepted	2.43794e-08	1.0468	1.2249	5e-09
E_-4	Screened	unavailable	—	—	2.5e-09
E_-2	Poor	6.28594e+11	1.0566	1.226	2.5e-09
E_+0	Poor	3.26683e+11	1.071	1.2094	2.5e-09
E_+2	Accepted	0.0021186	1.1235	1.2273	2.5e-09

Matching-source accounting

conditional-one-scale-mixed-scalar: 2; hierarchical-hard-plus-direct: 10; legacy-common-scale: 4

Neutrino observables — Small topology quartic / frozen UV

Numerical predictions are recorded only where the frozen campaign reached the neutrino evaluator. Blank results were screened out, not assigned artificial residuals.

Model	Delta m21^2	Delta m3l^2	sin2 th12	sin2 th13	sin2 th23
A_-4	—	—	—	—	—
A_-2	0.53048	27.794	0.12947	0.11384	0.63564
A_+0	128.56	14565	0.60478	0.40045	0.40957
B_-3	—	—	—	—	—
B_-1	7.537e-05	0.002511	0.3088	0.02248	0.47
B_+1	7.5371e-05	0.002511	0.3088	0.02248	0.47
C_-3	—	—	—	—	—
C_-1	7.537e-05	0.002511	0.3088	0.02248	0.47
C_+1	7.5371e-05	0.002511	0.3088	0.02248	0.47
D_-2	0.79926	1.6122	0.46178	0.079098	0.99758
D_+0	0.90356	2.8206	0.32147	0.097756	0.26349
D_+2	7.537e-05	0.002511	0.3088	0.02248	0.47
E_-4	—	—	—	—	—
E_-2	0.76548	1.5839	0.042422	0.12327	0.62346
E_+0	0.55198	1.1153	0.99155	0.20469	0.23757
E_+2	7.5394e-05	0.0025118	0.3088	0.02248	0.47

Large topology quartic / frozen UV

Saved low-energy endpoints; masses at muS (TeV). Missing entries are explicitly unavailable. B/C existing imported fits are not new bulk optimisations.

Model	status	chi2	mS1	mS2	lambdaT3
A_-4	Screened	unavailable	—	—	0.5
A_-2	Accepted	0.000132883	1.0547	1.151	0.5
A_+0	Accepted	5.73442	1.0691	1.1368	0.5
B_-3	Screened	unavailable	—	—	0.5
B_-1	Accepted	0.0329504	1.017	1.1654	0.5
B_+1	Accepted	3.60644e-08	1.0449	1.1646	0.5
C_-3	Screened	unavailable	—	—	0.5
C_-1	Accepted	4.58286e-11	1.017	1.1654	0.5
C_+1	Accepted	4.29412e-06	1.0449	1.1646	0.5
D_-2	Accepted	0.0567632	0.99384	1.225	0.5
D_+0	Accepted	0.000132648	1.0063	1.2083	0.5
D_+2	Accepted	4.22469	1.0489	1.2253	0.5
E_-4	Screened	unavailable	—	—	0.25
E_-2	Accepted	3.57434	1.0568	1.2261	0.25
E_+0	Accepted	1.47448	1.0712	1.2095	0.25
E_+2	Accepted	0.176217	1.1237	1.2274	0.25

Matching-source accounting

conditional-one-scale-mixed-scalar: 2; hierarchical-hard-plus-direct: 10; legacy-common-scale: 4

Neutrino observables — Large topology quartic / frozen UV

Numerical predictions are recorded only where the frozen campaign reached the neutrino evaluator. Blank results were screened out, not assigned artificial residuals.

Model	Delta m21^2	Delta m3l^2	sin2 th12	sin2 th13	sin2 th23
A_-4	—	—	—	—	—
A_-2	7.5364e-05	0.0025108	0.3088	0.02248	0.47
A_+0	7.5283e-05	0.0025091	0.30714	0.02239	0.50678
B_-3	—	—	—	—	—
B_-1	7.5215e-05	0.0025123	0.30892	0.022512	0.46987
B_+1	7.537e-05	0.002511	0.3088	0.02248	0.47
C_-3	—	—	—	—	—
C_-1	7.537e-05	0.002511	0.3088	0.02248	0.47
C_+1	7.5371e-05	0.002511	0.3088	0.02248	0.47
D_-2	7.5361e-05	0.0025109	0.3088	0.022478	0.46631
D_+0	7.5364e-05	0.0025108	0.3088	0.02248	0.47
D_+2	7.4906e-05	0.0025152	0.30999	0.022533	0.50066
E_-4	—	—	—	—	—
E_-2	7.5248e-05	0.002537	0.31168	0.022352	0.44971
E_+0	7.5151e-05	0.0025047	0.30841	0.022501	0.45216
E_+2	7.5413e-05	0.0025124	0.30872	0.022483	0.46361

Small topology quartic / running UV

Saved low-energy endpoints; masses at muS (TeV). Missing entries are explicitly unavailable. B/C existing imported fits are not new bulk optimisations.

Model	status	chi2	mS1	mS2	lambdaT3
A_-4	Screened	unavailable	—	—	5e-09
A_-2	Screened	unavailable	—	—	5e-09
A_+0	Screened	unavailable	—	—	5e-09
B_-3	Screened	unavailable	—	—	5e-09
B_-1	Accepted*	0.00288804	1.8389	2.9073	8.998e-06
B_+1	Screened	unavailable	—	—	5e-09
C_-3	Screened	unavailable	—	—	5e-09
C_-1	Accepted*	3.13301e-08	2.2303	3.2168	2.134e-05
C_+1	Screened	unavailable	—	—	5e-09
D_-2	Screened	unavailable	—	—	5e-09
D_+0	Screened	unavailable	—	—	5e-09
D_+2	Screened	unavailable	—	—	5e-09
E_-4	Screened	unavailable	—	—	2.5e-09
E_-2	Screened	unavailable	—	—	2.5e-09
E_+0	Screened	unavailable	—	—	2.5e-09
E_+2	Screened	unavailable	—	—	2.5e-09

Matching-source accounting

conditional-one-scale-mixed-scalar: 2; hierarchical-hard-plus-direct: 10; legacy-common-scale: 4

Neutrino observables — Small topology quartic / running UV

Numerical predictions are recorded only where the frozen campaign reached the neutrino evaluator. Blank results were screened out, not assigned artificial residuals.

Model	Delta m21^2	Delta m3l^2	sin2 th12	sin2 th13	sin2 th23
A_-4	—	—	—	—	—
A_-2	—	—	—	—	—
A_+0	—	—	—	—	—
B_-3	—	—	—	—	—
B_-1	7.5354e-05	0.0025114	0.30911	0.022484	0.47012
B_+1	—	—	—	—	—
C_-3	—	—	—	—	—
C_-1	7.537e-05	0.002511	0.3088	0.02248	0.47
C_+1	—	—	—	—	—
D_-2	—	—	—	—	—
D_+0	—	—	—	—	—
D_+2	—	—	—	—	—
E_-4	—	—	—	—	—
E_-2	—	—	—	—	—
E_+0	—	—	—	—	—
E_+2	—	—	—	—	—

Saved Yukawa information and missing outputs

Matrix entries are exported separately in tables/saved_yukawas.csv when an explicit local fitted_config.json or fit.json contains all 18 entries. The code does not infer Yukawas from low-energy observables, or substitute generic defaults for missing fit records. Yukawa sources located for this reporting environment: 18/48 cases. Unavailable cases remain blank in tables/yukawa_availability.csv.

RGE and oval figures

The saved CSV contains only endpoint diagnostics. It is insufficient to construct genuine UV/EFT1/final-C5 running functions of scale or the UV-parameter response required for sensitivity ovals. The existing historical HTML visualisations remain accessible via the interactive index with a historical label. No numerical integration was performed during this reporting build.

Reproducibility

The source and missing-data manifest is in provenance.json. This PDF and its plots are generated by scripts/BuildGeneralScalar48FinalReports.py using only existing frozen data. No fitting, matching or running is executed.